

**Evaluating Expressions with Radicals**Question Bank 4.05 **ANSWER KEY**

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*Show your work. Circle your final answer.***For questions 1 through 6, evaluate.**

1.  $\sqrt{81}$

**9**

2.  $\sqrt[3]{125}$

**5**

3.  $\sqrt[3]{-8}$

**-2***An odd root keeps the sign of its radicand.*

4.  $\sqrt{\frac{4}{25}}$

 **$\frac{2}{5}$** *The numerator and the denominator each get the root.*

5.  $\sqrt{5^2 + 11}$

**6***Finish under the bar:  $25 + 11 = 36$ .*

6.  $\sqrt[3]{3^3 + 37}$

**4** *$27 + 37 = 64$ .*

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a.  $\sqrt{100} \cdot 2^{-2}$

$$\frac{5}{2}$$

$$10 \cdot \frac{1}{4} \quad .$$

b.  $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

$$\frac{17}{4}$$

$$\frac{1}{4} + 4 \quad .$$

c.  $\sqrt[3]{-64} - 3^{-1}$

$$-\frac{13}{3}$$

$$-4 - \frac{1}{3} \quad .$$

8. Evaluate each expression.

a.  $\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$

$$0.2$$

$$3 - 1.4(2) \quad .$$

b.  $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

$$-\frac{17}{9}$$

$$\frac{1}{9} - 2 \quad .$$

c.  $\sqrt{4^3 - 39} \cdot 5^{-3}$

$$\frac{1}{25}$$

$$5 \cdot \frac{1}{125} \quad .$$

### Working backwards.

9. Find the value of  $n$  that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a.  $\sqrt{n} = 12$

$n = 144$

b.  $\sqrt[3]{n} = -5$

$n = -125$

c.  $\sqrt{n+7} = 9$

$n = 74$

$n + 7 = 81$  .

### Reading someone else's work.

10. Tonya writes  $\sqrt{16+9} = \sqrt{16} + \sqrt{9} = 7$  . Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

$\sqrt{16+9} = \sqrt{25} = 5$  . *The bar groups 16 + 9 into a single radicand, so the addition is finished before any root is taken.*

*Her method would need the root of a sum to equal the sum of the roots, and it does not.*

11. Which is the value of  $\sqrt{36+64}$  ?

☐ A. 14

☐ B. 100

☒ C. 10

☐ D. 50

**C**

12. Select ALL of the expressions whose value is 3.

☐ A.  $\sqrt{3^2+16}$

☒ B.  $\sqrt[3]{2^2+23}$

☒ C.  $\sqrt{4^2-7}$

☒ D.  $\sqrt[3]{27}$

☐ E.  $\sqrt[3]{-27}$

☒ F.  $\sqrt{9}$

**B, C, D and F**

*C's radicand is  $16-7=9$  and B's is  $4+23=27$  .*